Q#	Type	Question	Choices
1	MCQ	What is the core difference between VMs and containers?	A) Containers virtualize hardware; VMs virtualize OS B) VMs virtualize hardware; containers virtualize OS-level resources C) Containers require more resources than VMs D) VMs start faster than containers
2	MCQ	Which kernel feature isolates what a container can “see” (e.g., PIDs, routes, IPs)?	A) Cgroups B) Namespaces C) OverlayFS D) Hypervisors
3	MCQ	Cgroups are responsible for which of the following?	A) Isolating network interfaces B) Controlling resource usage (CPU, memory, I/O) C) Creating container images D) Scheduling Pods
4	MCQ	Containers are lightweight mainly because they share:	A) Kernel and system libraries B) Full OS instances C) Hardware virtualization layer D) Hypervisor
5	MCQ	What does the PID namespace isolate?	A) Processes within a container B) Network interfaces C) Filesystem D) Memory limits
6	MCQ	Containers run as:	A) Full VMs B) Standard processes on the host OS C) Hypervisor processes D) Detached kernels
7	MCQ	Which feature provides filesystem isolation?	A) Cgroups B) Namespaces C) OverlayFS D) Scheduler
8	MCQ	Which of the following does not isolate containers?	A) Namespaces B) Cgroups C) Docker daemon D) Hypervisor
9	MCQ	Containers differ from VMs because they:	A) Emulate full hardware B) Share host OS kernel C) Take longer to start D) Require a separate OS
10	MCQ	Why do containers start faster than VMs?	A) They emulate hardware B) They use shared host OS kernel C) They require booting a separate kernel D) They rely on full OS installation
11	MCQ	What does the UTS namespace isolate?	A) Hostname and domain name B) CPU resources C) Filesystem D) Network routes
12	MCQ	In Linux containers, which mechanism controls resource limits?	A) Namespaces B) Cgroups C) Dockerfile D) OverlayFS
13	MCQ	Which layer of OverlayFS do containers write to?	A) Lower layer B) Upper/writable layer C) Base image layer D) Shared library layer
14	MCQ	Container images primarily use:	A) Writable layers B) Read-only layers C) Hypervisor images D) Kernel snapshots
15	MCQ	Docker Hub is mainly used for:	A) Running containers B) Storing and sharing images C) Controlling resources D) Scheduling Pods
16	MCQ	Which statement describes Docker images?	A) Fully isolated VMs B) Stack of read-only layers C) Hardware emulation D) Single executable file

17	MCQ	Which Docker network driver provides the highest isolation?	A) Bridge B) Host C) None D) Overlay
18	MCQ	What mechanism connects container interfaces to Docker's bridge?	A) veth pair B) Namespaces C) Cgroups D) OverlayFS
19	MCQ	Docker DNS resolves container names only on which network?	A) Default bridge B) Host network C) User-defined bridge D) None network
20	MCQ	Docker images consist of:	A) Single layer B) Multiple stacked layers C) Writable base only D) Kernel snapshots
21	MCQ	A container using the "none" network driver receives:	A) Full network access B) No network C) NAT-ed bridge network D) Host IP
22	MCQ	The writable layer of a container is called:	A) Upper layer B) Lower layer C) Base image D) Snapshot
23	MCQ	The veth pair exists to:	A) Share filesystem B) Connect container network interface to bridge C) Limit CPU D) Mount images
24	MCQ	OverlayFS benefits images by enabling:	A) Writable layers only B) Copy-on-write and sharing base layers C) Direct kernel access D) Network isolation
25	MCQ	Which component runs containers on Kubernetes nodes?	A) kubelet B) API server C) Scheduler D) Controller-manager
26	MCQ	What does etcd store?	A) Cluster state and configuration B) Container images C) Network routing D) Node logs
27	MCQ	The scheduler decides placement based on:	A) Resource availability and constraints B) Image size C) Pod readiness D) Network latency
28	MCQ	Which component applies the desired state?	A) Controller-manager B) kubelet C) Scheduler D) etcd
29	MCQ	The API server functions as:	A) Gateway for client requests B) Scheduler C) Node controller D) Volume provisioner
30	MCQ	Worker nodes typically run:	A) kubelet and container runtime B) API server C) Scheduler D) etcd
31	MCQ	The control plane enforces:	A) Desired cluster state B) Node storage C) Pod networking only D) Image registry
32	MCQ	Horizontal Pod Autoscaler (HPA) scales based on:	A) Metrics like CPU or memory usage B) Number of nodes C) ReplicaSet size D) Pod logs
33	MCQ	Vertical Pod Autoscaler (VPA) adjusts:	A) Resource requests and limits B) Node size C) Pod count D) Storage volume
34	MCQ	Which component detects unreachable nodes?	A) Node controller B) Scheduler C) kubelet D) API server

35	MCQ	The node controller responds when:	A) Node is down or unhealthy B) Pod fails C) Deployment scales D) ConfigMap updates
36	MCQ	Which Kubernetes object holds cluster-wide configuration?	A) ConfigMap B) Secret C) Deployment D) Service
37	MCQ	What does a Pod represent in Kubernetes?	A) Single container B) Group of containers scheduled together C) Node D) Deployment
38	MCQ	Containers in a Pod share:	A) Network namespace and storage B) CPU resources only C) Host kernel D) Hypervisor layer
39	MCQ	The pause container is used to:	A) Initialize Pod networking B) Run application logic C) Mount volumes only D) Provide persistent storage
40	MCQ	Why do containers inside a Pod communicate via localhost?	A) They share network namespace B) They share PID namespace only C) Kubernetes NATs the traffic D) They share separate IPs
41	MCQ	Pod networking in Kubernetes is based on:	A) Flat network model B) NAT per Pod C) Host-only networking D) OverlayFS only
42	MCQ	Which Service type assigns a virtual internal IP?	A) ClusterIP B) NodePort C) LoadBalancer D) ExternalName
43	MCQ	Pods are ephemeral because:	A) They can be terminated and recreated B) They use persistent storage C) They are VMs D) They are always long-running
44	MCQ	ReplicaSets ensure:	A) Desired number of Pod replicas are running B) Node availability C) Persistent storage D) Network isolation
45	MCQ	A Pod with two containers shares:	A) Network namespace B) Filesystem only C) Host kernel only D) CPU limits only
46	MCQ	The purpose of sidecar containers is:	A) Extend Pod functionality B) Main application C) Kubernetes control D) Node monitoring only
47	MCQ	Pods communicate without NAT because Kubernetes assumes:	A) Flat network B) Pod-specific NAT C) Host network only D) OverlayFS routing
48	MCQ	A Pod is the smallest:	A) Deployable unit in Kubernetes B) Node C) ReplicaSet D) Cluster component
49	MCQ	VMs differ from containers primarily because:	A) VMs virtualize OS B) VMs virtualize hardware C) Containers emulate hardware D) Containers are slower to start
50	MCQ	Which component is required for VMs but not containers?	A) Hypervisor B) Cgroups C) Namespace D) Docker Engine
51	MCQ	Type-1 hypervisors run:	A) Directly on hardware B) On top of host OS C) Inside a container D) Only Linux OS
52	MCQ	Containers start faster than VMs because they:	A) Require less I/O B) Share host OS kernel C) Use virtualization D) Boot a separate kernel

53	MCQ	Hypervisors mainly manage:	A) CPU, memory, and storage virtualization B) Network namespace C) OverlayFS layers D) Pod lifecycle
54	MCQ	Containers share:	A) Host OS kernel B) Hypervisor C) CPU only D) Node IP only
55	MCQ	Which namespace ensures process isolation?	A) PID B) NET C) MNT D) UTS
56	MCQ	Cgroups provide:	A) Resource limits B) Filesystem isolation C) Process isolation D) Pod scheduling
57	MCQ	VMs emulate:	A) Full hardware B) Kernel only C) Processes D) Network namespace
58	MCQ	Containers provide isolation through:	A) Namespaces and Cgroups B) Hypervisor C) Full OS D) VMs
59	MCQ	Which best describes a VM?	A) Full hardware emulation B) Containerized process C) Namespace-only isolation D) Sidecar container
60	MCQ	Hypervisor overhead exists because:	A) It emulates hardware B) Containers share kernel C) Namespaces limit CPU D) OverlayFS duplicates data
61	MCQ	Docker default bridge provides:	A) NAT-ed network for containers B) Host network C) Overlay networking D) None
62	MCQ	Kubernetes flat network model ensures:	A) All Pods can communicate without NAT B) NodePort only C) ClusterIP only D) Separate subnet per Pod
63	MCQ	kube-proxy rewrites packets to support:	A) Service load balancing B) DNS resolution C) OverlayFS mounting D) Sidecar container initialization
64	MCQ	VXLAN is used for:	A) Overlay networking across nodes B) Process isolation C) Resource limits D) Volume mounts
65	MCQ	In Docker, NAT occurs when:	A) Containers use default bridge B) Containers use host network C) Containers use none D) Pods share network namespace
66	MCQ	Container veth interfaces connect:	A) Containers to bridges B) CPU to memory C) Namespaces D) Hypervisor to kernel
67	MCQ	Kubernetes Services use which mechanism?	A) kube-proxy B) OverlayFS C) Dockerfile D) Cgroups
68	MCQ	ClusterIP provides:	A) Internal virtual IP B) External IP C) NodePort D) LoadBalancer
69	MCQ	NodePort exposes:	A) Service on host port B) Cluster-internal IP C) Overlay network D) Pod IP only
70	MCQ	Pod-to-Pod traffic across nodes uses:	A) Overlay network or CNI plugin B) Host network only C) NAT D) Sidecar container
71	MCQ	Docker's default DNS works only on:	A) Default bridge network B) Host network C) Overlay network D) None network
72	MCQ	Kubernetes networking is fundamentally:	A) Flat, NAT-less B) NAT per Pod C) Host-only D) VM-based
73	MCQ	Deployments manage:	A) ReplicaSets and Pods B) Nodes only C) Network policies D) Volumes only

74	MCQ	ReplicaSets maintain:	A) Desired number of Pods B) Node availability C) Persistent volumes D) Overlay networks
75	MCQ	Node Autoscaler triggers when:	A) Resource demand exceeds cluster capacity B) Pod fails C) Deployment scales D) ClusterIP service is unavailable
76	MCQ	ClusterIP Services provide:	A) Internal access only B) External access only C) NodePort D) LoadBalancer
77	MCQ	HPA depends on:	A) Metrics (CPU/memory) B) ReplicaSet size C) Node availability D) Pod labels
78	MCQ	Ingress works at which layer?	A) L7 (HTTP/HTTPS) B) L4 (TCP/UDP) C) L3 (IP routing) D) L2 (Ethernet)
79	MCQ	Which Kubernetes object is declarative?	A) Deployment B) Pod only C) Node D) ServiceAccount
80	MCQ	Which component handles rollout/ rollback?	A) Deployment controller B) Scheduler C) kubelet D) etcd
81	MCQ	Which resource ensures Pods run on all nodes?	A) DaemonSet B) Deployment C) ReplicaSet D) StatefulSet
82	MCQ	Which object models a running application?	A) Deployment B) Pod C) ReplicaSet D) Service
83	MCQ	A Deployment with 3 replicas ensures:	A) 3 Pods running B) 3 nodes active C) 3 Containers per Pod D) 3 volumes mounted
84	MCQ	Autoscaling decisions depend on:	A) Metrics and thresholds B) Node IP C) Pod PID namespace D) OverlayFS layers
85	MCQ	Which Kubernetes feature allows multiple containers to share the same lifecycle?	A) Pod B) Deployment C) ReplicaSet D) Node
86	MCQ	Which Docker storage driver uses copy-on-write for layers?	A) OverlayFS B) AUFS C) Device Mapper D) Btrfs
87	MCQ	In Kubernetes, a StatefulSet is used for:	A) Stateful applications B) Stateless applications C) Horizontal scaling only D) Node management
88	MCQ	Which type of Kubernetes Service exposes an application externally via cloud provider load balancer?	A) LoadBalancer B) ClusterIP C) NodePort D) ExternalName
89	MCQ	What does the kube-scheduler primarily consider when scheduling Pods?	A) Node resources, affinity/anti-affinity, taints/tolerations B) Network latency C) OverlayFS layers D) Image size
90	MCQ	In Docker, which command builds an image from a Dockerfile?	A) docker build B) docker run C) docker commit D) docker push

91	MCQ	Which Kubernetes object is declarative and manages ReplicaSets?	A) Deployment B) Pod C) StatefulSet D) Service
92	MCQ	Kubernetes PersistentVolume (PV) is:	A) Storage resource in the cluster B) Compute resource C) Network namespace D) Container image
93	MCQ	Horizontal Pod Autoscaler (HPA) scales Pods based on:	A) CPU/memory metrics B) Number of nodes C) Pod names D) OverlayFS usage
94	MCQ	Which namespace isolates hostname and domain name in Linux containers?	A) UTS B) PID C) NET D) MNT
95	MCQ	Kubernetes ConfigMap is used for:	A) Storing configuration data B) Storing secrets only C) Scheduling Pods D) Resource limits
96	MCQ	Which CNI plugin type is commonly used for overlay networking in Kubernetes?	A) Flannel B) OverlayFS C) veth D) kube-proxy
97	MCQ	A sidecar container is typically used to:	A) Extend main container functionality B) Replace the main container C) Control the host kernel D) Emulate hardware
98	MCQ	What command lists all running Docker containers?	A) docker ps B) docker images C) docker run D) docker network ls
99	MCQ	Kubernetes DaemonSet ensures:	A) A Pod runs on all nodes B) Deployment scaling C) Persistent storage mounting D) Overlay network creation
100	MCQ	The pause container in a Pod:	A) Holds network namespace B) Runs main application C) Manages resource limits D) Updates Kubernetes API
101	MCQ	Docker Hub is primarily used to:	A) Store and share images B) Run containers C) Apply resource limits D) Manage Kubernetes clusters
102	MCQ	In OverlayFS, the upper layer is:	A) Writable B) Read-only C) Base image D) Kernel layer
103	MCQ	Which Kubernetes object defines desired state and allows declarative updates?	A) Deployment B) Pod C) Node D) ServiceAccount
104	MCQ	Vertical Pod Autoscaler (VPA) adjusts:	A) Resource requests and limits B) Pod count C) Node IPs D) Overlay network bridges
105	MCQ	Docker veth pair is used to:	A) Connect container interface to bridge B) Limit CPU C) Mount volumes D) Share PID namespace
106	MCQ	Kubernetes ClusterIP Service provides:	A) Internal virtual IP only B) External load balancer C) NodePort D) StatefulSet IP

107	MCQ	NodePort Service exposes:	A) Service on a node port B) Internal cluster IP only C) Overlay network IP D) None
108	MCQ	Which Kubernetes object is used for rolling updates?	A) Deployment B) Pod C) StatefulSet D) ConfigMap
109	MCQ	The Kubernetes control plane component that enforces desired state:	A) Controller-manager B) kubelet C) Scheduler D) etcd
110	MCQ	ReplicaSet ensures:	A) Desired number of Pods B) Node scaling C) Overlay networking D) Persistent storage
111	MCQ	In Docker, the COPY instruction in a Dockerfile:	A) Copies files into image B) Copies container runtime C) Mounts volumes D) Creates a network namespace
112	MCQ	Kubernetes StatefulSet differs from Deployment because:	A) Stable network ID and persistent storage B) Stateless C) Only for single node D) Uses Docker Hub only
113	Explai	Explain how namespaces isolate containers at the OS level.	
114	Explai	Describe why containers start faster and use fewer resources than VMs.	
115	Explai	Explain why containers are processes, not virtual machines.	
116	Explai	Explain how namespaces contribute to multi-tenant security.	
117	Explai	Describe the relationship between containerized applications and the host OS kernel.	
118	Explai	Explain how Cgroups protect the host from resource exhaustion.	
119	Explai	Explain why Docker images use read-only layers and how copy-on-write works.	

120	Explai	Describe how OverlayFS reduces duplicated data across containers.	
121	Explai	Explain the relationship between Docker Engine, container images, and containers.	
122	Explai	Explain how Docker embedded DNS works on a user-defined bridge.	
123	Explai	Describe how Docker networking uses NAT in the default bridge.	
124	Explai	Explain why Docker containers do not require full OS installations.	
125	Explai	Explain how the Kubernetes control plane enforces desired state.	
126	Explai	Describe the role of kubelet.	
127	Explai	Explain the function of the scheduler.	
128	Explai	Describe why etcd is critical for Kubernetes reliability.	
129	Explai	Explain how controller-manager maintains cluster health.	
130	Explai	Describe the responsibilities of control-plane components.	
131	Explai	Explain what a Pod is and why multiple containers may run in one Pod.	
132	Explai	Describe why Pods are ephemeral and how ReplicaSets maintain availability.	
133	Explai	Explain how the pause container stabilizes networking.	

134	Explai	Describe the sidecar design pattern.	
135	Explai	Explain why Pods—not containers—are the scheduling unit.	
136	Explai	Describe how Pod-to-Pod communication works.	
137	Explai	Compare container and VM resource models.	
138	Explai	Explain the purpose of hypervisors.	
139	Explai	Describe how OS virtualization differs from hardware virtualization.	
140	Explai	Explain why VMs provide stronger isolation than containers.	
141	Explai	Describe how VM boot speed compares to containers and why.	
142	Explai	Explain when VMs may be required instead of containers.	
143	Explai	Explain the purpose of kube-proxy.	
144	Explai	Describe how Docker default bridge networking works.	
145	Explai	Explain why Kubernetes requires NAT-less Pod networking.	
146	Explai	Describe Pod networking across nodes.	
147	Explai	Explain overlay networks in container clusters.	
148	Explai	Explain the role of veth pairs.	
149	Explai	Explain the difference between HPA and VPA.	

150	Explai	Describe why Deployments simplify version rollout.	
151	Explai	Explain how multi-replica deployments achieve high availability.	
152	Explai	Describe the purpose of Ingress.	
153	Explai	Explain the role of Services in connecting Pods.	
154	Explai	Describe how cluster autoscalers interact with node resources.	
155	Explai	Explain why Pods are the smallest deployable unit in Kubernetes.	
156	Explai	Describe how namespaces provide isolation in containers.	
157	Explai	Explain how Cgroups prevent resource exhaustion in a multi-tenant environment.	
158	Explai	Describe the difference between overlay and bridge networks in Docker.	
159	Explai	Explain how the Kubernetes scheduler determines Pod placement.	
160	Explai	Describe the function of the Kubernetes controller-manager.	
161	Explai	Explain the difference between HPA and VPA in scaling Pods.	
162	Explai	Describe how sidecar containers extend Pod functionality.	
163	Explai	Explain how Docker copy-on-write improves storage efficiency.	

164	Explai	Describe how ConfigMaps and Secrets differ in Kubernetes.	
165	Explai	Explain the purpose of DaemonSets in Kubernetes clusters.	
166	Explai	Describe the process of rolling updates using a Deployment.	
167	Explai	Explain why Kubernetes requires a flat, NAT-less Pod network.	
168	Explai	Describe how overlay networking allows Pods to communicate across nodes.	
169	Desigr	Design a containerization plan for a legacy application running on a VM.	
170	Desigr	Design a setup where two tightly coupled services run in the same containerized environment and justify your choice.	
171	Desigr	Design a multi-container Docker environment where services communicate securely.	
172	Desigr	Design a Docker-based microservice using separate images with shared base layers.	
173	Desigr	Design a Deployment with auto-scaling and explain which components enforce scaling.	
174	Desigr	Design an HA cluster layout using multi-master control plane redundancy.	

175	Desigr	Design a microservice where two containers must share localhost.	
176	Desigr	Design a sidecar-based logging solution for an application.	
177	Desigr	Design a migration strategy from VMs to containers.	
178	Desigr	Design a hybrid system using both VMs and containers.	
179	Desigr	Design a service exposure strategy using ClusterIP + Ingress.	
180	Desigr	Design a multi-node Pod network architecture using overlay networking.	
181	Desigr	Design a 3-tier application using Deployments and Services.	
182	Desigr	Design a scalable stateless application with HPA.	
183	Desigr	Design a Kubernetes Deployment for a web application with 3 replicas and auto-scaling.	
184	Desigr	Design a Docker Compose setup for two microservices communicating over a user-defined bridge.	
185	Desigr	Design a sidecar logging solution for a containerized application.	
186	Desigr	Design a hybrid system using both VMs and containers for legacy and new applications.	

187	Desigr	Design a persistent storage solution for a StatefulSet managing a database.	
188	Desigr	Design a multi-node Kubernetes overlay network for Pod-to-Pod communication.	
189	Desigr	Design a CI/CD pipeline using Docker and Kubernetes for automatic rolling updates.	
190	Desigr	Design a monitoring and alerting solution for containerized applications in Kubernetes.	